

1. Identify any five real-world applications or online services that involve communication, authentication, or transactions. For each application, investigate and identify:

The security feature/service required, such as confidentiality, integrity, authentication, authorization, availability

The security mechanism used to provide the identified security feature.

The security protocol or technology used to implement the mechanism.

Briefly explain how the mechanism/protocol provides the identified security feature.

Security is important in real-world applications because they handle personal information, communication, financial transactions and important files. Different security features are used to protect this information from unauthorized access, modification and attacks.

following table shows five real-world applications and the security mechanisms and technologies they use

Application	Security Feature	Security Mechanism	Protocol / Technology	How it provides security
Gmail	Authentication	Password + Multi-Factor Authentication (MFA)	OAuth 2.0, MFA	Verifies the identity of the user before allowing access to the account.
WhatsApp	confidentiality	End-to-End Encryption	Signal Protocol	Encrypts messages so that only the sender and intended receiver can read them.
Online Banking	Confidentiality	Encryption	HTTPS / TLS	Encrypts information exchanged between the user's device and the bank server.

Google Drive	Authorization	access Control	OAuth 2.0	Allows users to control who can access their files and what permissions they have.
Amazon	Integrity	Secure communication and transaction protection	HTTPS / TLS	Protects transaction information from being modified while it is being transmitted.

Gmail:

Gmail uses several security features to protect user accounts. Passwords are used for authentication, while Multi-Factor Authentication provides an additional layer of security. OAuth 2.0 is also used for secure authorization when applications access Google services.

Security Feature: Authentication

Mechanism: Password + MFA

Technology: OAuth 2.0, MFA

It verifies the identity of the user and helps prevent unauthorized people from accessing the account.

WhatsApp:

WhatsApp uses end-to-end encryption to protect personal communication. Messages are encrypted before they leave the sender's device and can only be decrypted by the intended recipient.

Security Feature: Confidentiality

Mechanism: End-to-End Encryption

Technology: Signal Protocol

This prevents third parties from reading the content of messages while they are being transmitted.

Online Banking:

Online banking applications handle sensitive information such as account details, passwords and financial transactions. HTTPS and TLS are used to protect communication between the user and the banking server.

Security Feature: Confidentiality

Mechanism: Encryption

Technology: HTTPS / TLS

TLS encrypts the data exchanged between the user's device and the bank server, making it difficult for attackers to read the information during transmission.

Google Drive:

Google Drive allows users to store and share files online. Access control is used to decide who can view, edit or share a file.

Security Feature: Authorization

Mechanism: Access Control

Technology: OAuth 2.0

The access control mechanism ensures that only users with the required permission can access or modify the files.

Amazon:

Amazon handles online shopping and payment transactions. Secure communication is required to protect customer and transaction information.

Security Feature: Integrity

Mechanism: Secure Communication and Transaction Protection

Technology: HTTPS / TLS

HTTPS/TLS protects information while it is being transmitted and helps prevent attackers from modifying the data during communication.

2. For any five real-world applications, investigate the security controls used to protect users and data. Classify the controls based on whether they provide Confidentiality, Integrity, or Availability, and determine how these controls help in preventing, detecting, and recovering from security attacks. Mention the mechanisms and protocols involved.

Security controls are measures used by applications to protect users, data and services from cyber attacks. These controls can provide **Confidentiality, Integrity and Availability (CIA)**.

- **Confidentiality:** Prevents unauthorized people from accessing information.
- **Integrity:** Ensures that information is not changed without authorization.

- **Availability:** Ensures that services and information are available when required.

table shows security controls used by different applications:

Security Controls and CIA Classification

Application	Security Control	CIA	Mechanism / Protocol	Prevention	Detection	Recovery
Gmail	Multi-Factor Authentication and Login Alerts	Confidentiality	MFA, OAuth 2.0	Prevents unauthorized users from logging in.	Login alerts can identify suspicious login attempts.	Account recovery helps the user regain access.
WhatsApp	End-to-End Encryption	Confidentiality	Signal Protocol	Prevents unauthorized people from reading messages.	Security notifications can indicate changes in security keys.	Account re-verification helps restore the account.
Online Banking	Encryption and Transaction Authentication	Confidentiality, Integrity	HTTPS/TLS, OTP, MFA	Protects account and transaction information.	Fraud monitoring can detect suspicious transactions.	Banks can block accounts and investigate fraudulent transactions.
Google Drive	Access Control and File Recovery	Confidentiality, Availability	OAuth 2.0, Access Permissions	Prevents unauthorized users from accessing files.	Activity records can help identify unauthorized changes.	Deleted files can be restored using recovery features.

Amazon	Secure Communication and Fraud Detection	Confidentiality, Integrity	HTTPS/TLS, MFA, Fraud Detection	Protects account and payment information.	Detects unusual login and transaction activity.	Account recovery and transaction dispute processes help users recover from incidents.
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Gmail Security Controls:

Gmail uses security controls such as Multi-Factor Authentication, login alerts and account recovery.

Confidentiality is provided by authentication controls that prevent unauthorized people from accessing emails.

Prevention: MFA makes unauthorized login more difficult.

Detection: Login alerts can inform users about suspicious activity.

Recovery: Account recovery mechanisms help users regain access if their account is compromised.

WhatsApp Security Controls:

WhatsApp uses end-to-end encryption to protect communication between users.

Confidentiality is the main security objective because messages should only be readable by the intended participants.

Prevention: Encryption prevents unauthorized parties from reading messages.

Detection: Security notifications can alert users when a security key changes.

Recovery: Users can re-verify their accounts when changing or restoring devices.

Online Banking Security Controls:

Online banking systems use several security controls because they handle financial information.

HTTPS/TLS protects communication, while OTP and MFA provide additional authentication.

Prevention: Encryption and authentication protect accounts and transactions from unauthorized access.

Detection: Banks use transaction monitoring and fraud detection to identify suspicious activity.

Recovery: Banks can block accounts, investigate transactions and help customers recover from fraudulent activity.

Google Drive Security Controls:

Google Drive uses access permissions and authentication to protect stored files.

Prevention: Access control prevents unauthorized users from viewing or modifying files.

Detection: Activity information can help identify changes made to files.

Recovery: Deleted files can often be recovered using the Trash and other recovery mechanisms.

Amazon Security Controls:

Amazon uses security controls to protect customer accounts, shopping activity and payment information.

HTTPS/TLS protects communication and additional authentication mechanisms help secure accounts.

Prevention: Encryption and authentication reduce the risk of unauthorized access and data theft.

Detection: Suspicious login and transaction activity can be identified using security and fraud detection systems.

Recovery: Account recovery and transaction dispute mechanisms help users deal with security incidents.